Network Intrusion Detection System using Snort

1. Introduction

A Network Intrusion Detection System (NIDS) is a cybersecurity tool used to monitor network traffic and detect suspicious or malicious activities. It helps security professionals identify potential threats, attacks, and unauthorized access attempts in a network environment.

This project focuses on implementing a Network Intrusion Detection System using Snort in Kali Linux to monitor network traffic and generate alerts for suspicious activity.

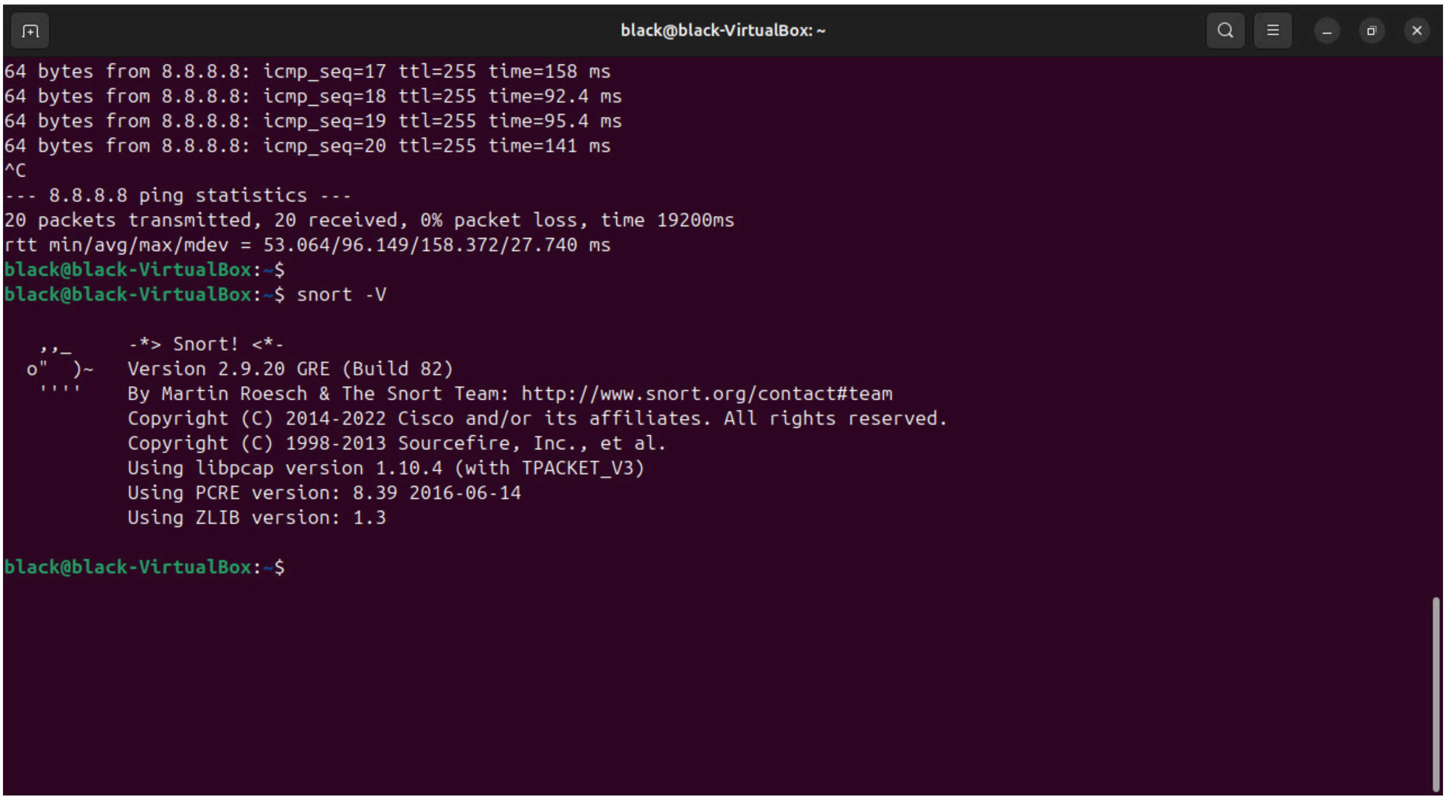
2. Objective

The main objective of this project is:

- To understand the basics of intrusion detection systems
- To install and configure Snort
- To monitor network traffic in real time
- To create custom alert rules
- To detect suspicious network activities

3. Tools and Technologies Used

- Snort
- Kali Linux Operating System
- Terminal / Command Line Interface
- Network Monitoring Tools


black@black-VirtualBox: ~
64 bytes from 8.8.8.8: icmp_seq=17 ttl=255 time=158 ms
64 bytes from 8.8.8.8: icmp_seq=18 ttl=255 time=92.4 ms
64 bytes from 8.8.8.8: icmp_seq=19 ttl=255 time=95.4 ms
64 bytes from 8.8.8.8: icmp_seq=20 ttl=255 time=141 ms
^C
--- 8.8.8.8 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19200ms
rtt min/avg/max/mdev = 53.064/96.149/158.372/27.740 ms
black@black-VirtualBox:~$
black@black-VirtualBox:~$ snort -V
-*> Snort! <*-
Version 2.9.20 GRE (Build 82)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.4 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.3
black@black-VirtualBox:~$

4. Snort Installation

Snort was installed in Kali Linux using the package manager.

Installation Command

sudo apt update

sudo apt install snort -y

Version Check

snort -V

The successful installation was verified by checking the Snort version.

5. Network Configuration

The available network interfaces were identified using the following command:

ip a

The network interface was selected for monitoring network traffic.

Example interfaces:

- wlan0
- eth0

6. Snort Configuration and Monitoring

Snort was configured to monitor network traffic using the following command:

sudo snort -A console -i wlan0 -c /etc/snort/snort.conf

This enabled Snort to analyze network packets in real time.

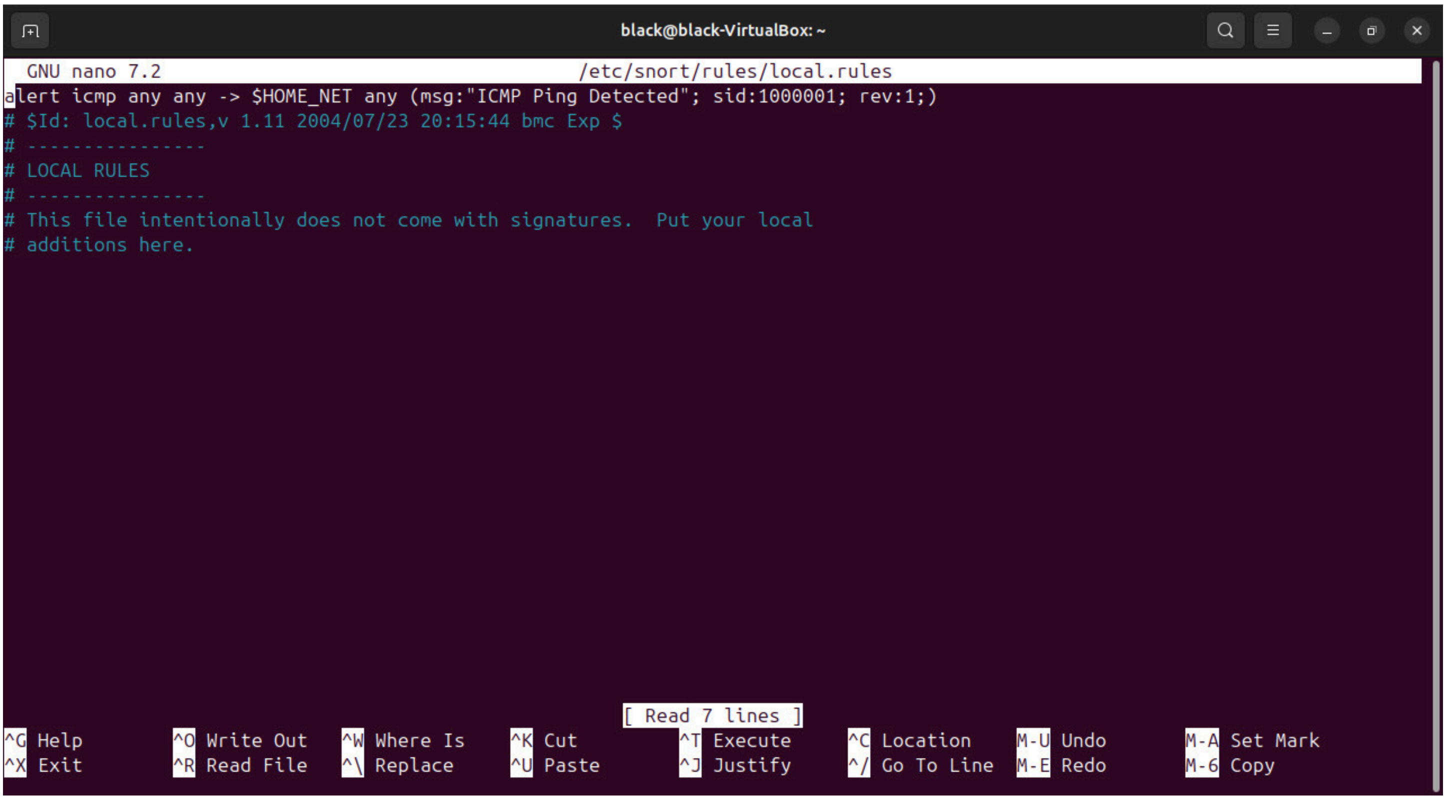
7. Custom Rule Creation

A custom rule was created to detect ICMP ping traffic.

Rule Used

alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001;)

This rule generates an alert whenever ping traffic is detected in the network



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8. Traffic Generation and Alert Detection

To test the intrusion detection system, network traffic was generated using the ping command.

Traffic Generation Command

ping 8.8.8.8

Snort successfully detected ICMP traffic and generated alerts in the terminal.

Example alert:

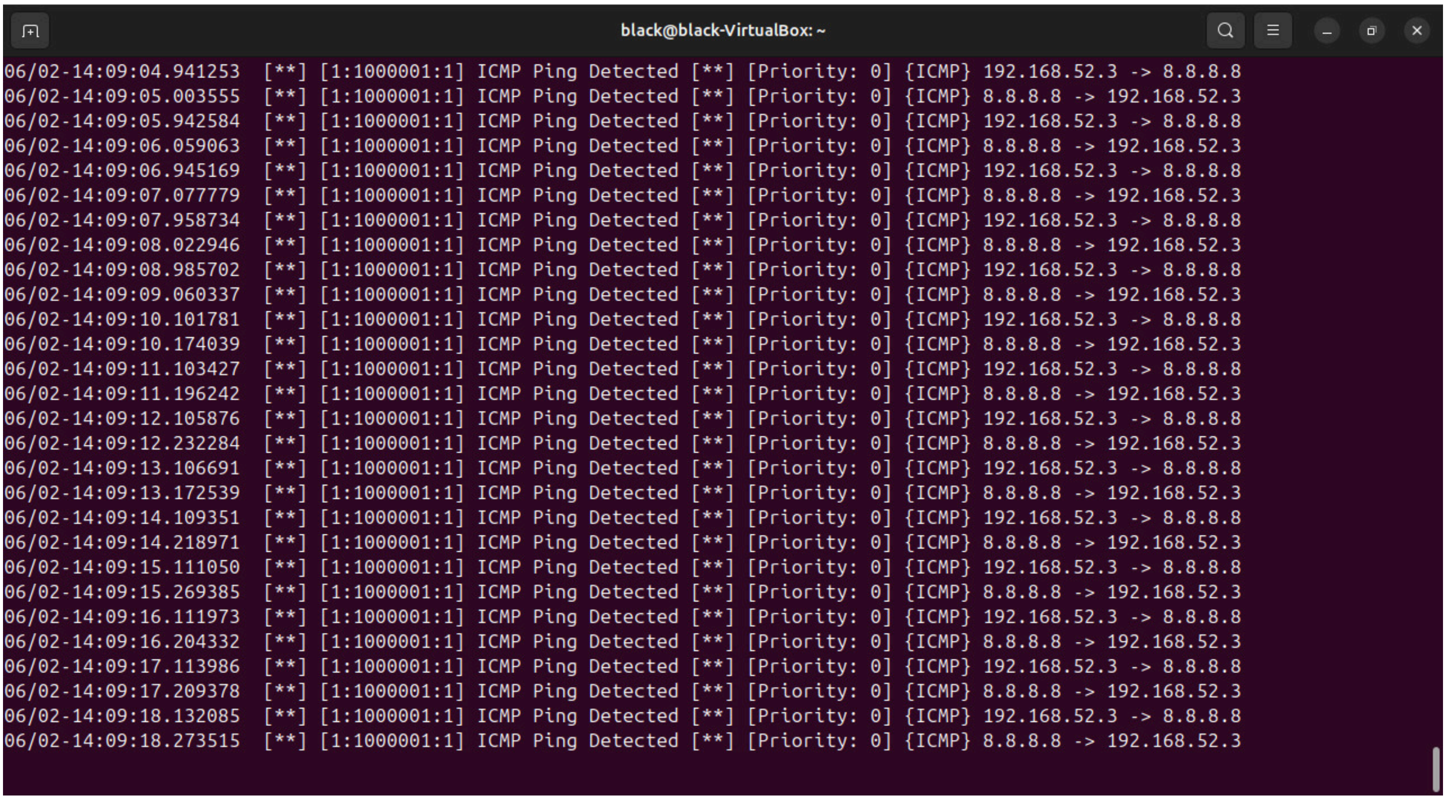
[**] ICMP Ping Detected [**]

This confirmed that the Network Intrusion Detection System was functioning correctly.

9. Output

The Snort intrusion detection system successfully monitored network traffic and generated alerts based on the custom detection rule.

The project demonstrated the ability of Snort to identify suspicious traffic and improve network monitoring



10. Learning Outcome

Through this project, I learned:

- **Basics of Network Intrusion Detection Systems (NIDS)**
- **Snort installation and configuration**
- **Real-time network traffic monitoring**
- **Custom rule creation in Snort**
- **Network security and threat detection**

11. Conclusion

This project provided practical experience in implementing a Network Intrusion Detection System using Snort. By monitoring network traffic and detecting ICMP ping activity, the project demonstrated how intrusion detection systems help improve cybersecurity and network security.

The implementation of Snort helped in understanding network monitoring techniques and threat detection mechanisms in a real-world environment.